IIA Code Setup

Table of Contents

1. IIA Code Setup: 2

IIA Code: 2

Run IIA: 2

1. Mongodb back up and restore 3

BackUp: 3

Restore: 4

1. IIA Dataset Setup 4

Uploading dataset for training in IIA: 4

Steps to add Records: 4

Steps to select the Training Parameters: 6

Steps to upload supporting documents in Assignment module : 8

3. Service Now Setup 13

4. IIA Deployment 17

Build Config: 17

Deploy build file: 18

5. IIA as Service 18

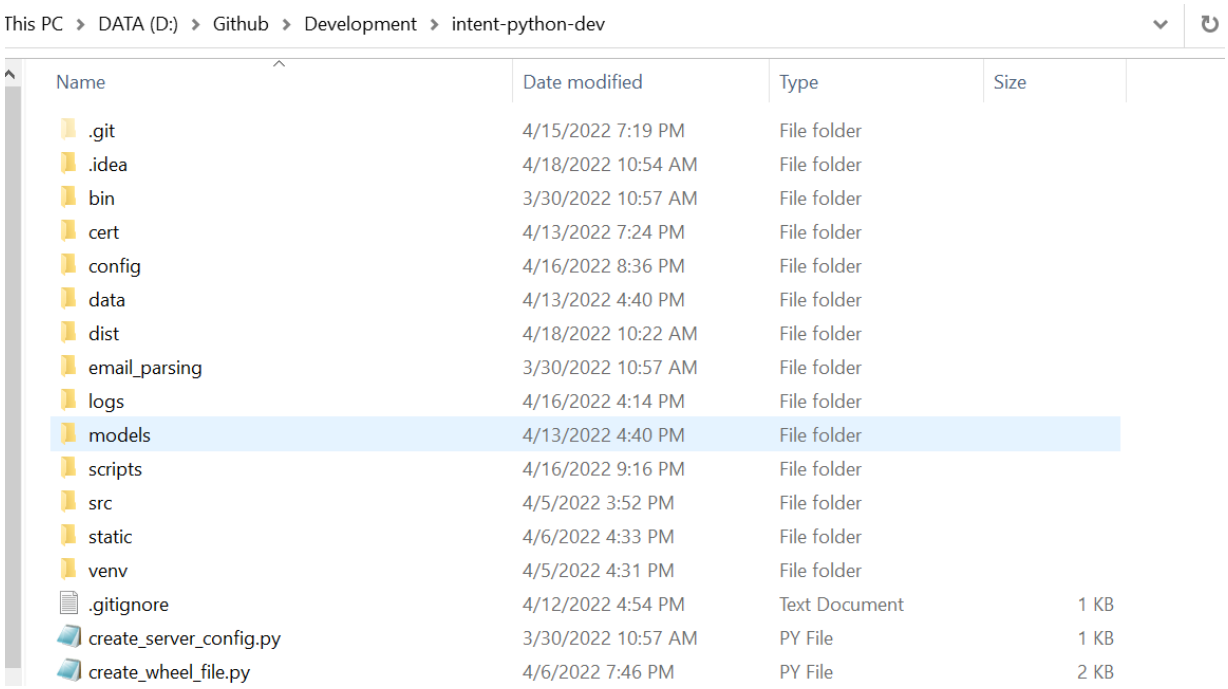
1. IIA Code Setup:

IIA Code:

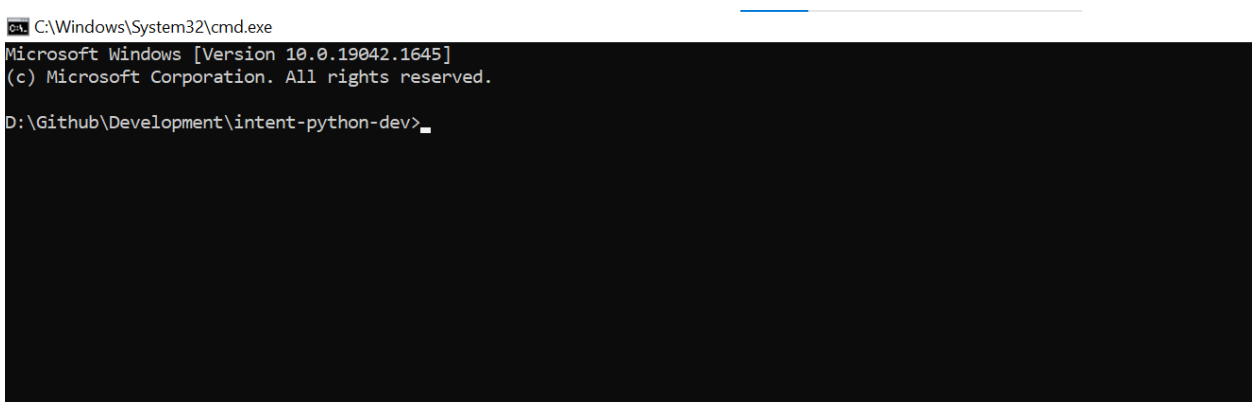
https://github.com/Infosys/Infosys-Intelligent-Assistant.git

Run IIA:

Navigate to the place where IIA back end code is present



Open Command prompt from the navigated location



1. If you are setting things up for the first time then issue

   python -m venv venv

   a folder named venv will be created

2. Activate the virtual environment

   Cd venv

   Cd scripts

   Activate

Cd..

Cd..

Pip install -r requirements.txt

Cd bin & run

```
D:\Github\Development\intent-python-dev>cd python -m venv env
The system cannot find the path specified.

D:\Github\Development\intent-python-dev>python -m venv env

D:\Github\Development\intent-python-dev>cd env

D:\Github\Development\intent-python-dev\env>cd Scripts

D:\Github\Development\intent-python-dev\env\Scripts>activate

(env) D:\Github\Development\intent-python-dev\env\Scripts>cd..

(env) D:\Github\Development\intent-python-dev\env>cd..

(env) D:\Github\Development\intent-python-dev>pip install -r requirements.txt
```

Follow the activate virtual environment and run bin going forward

1. Mongodb back up and restore

BackUp:

mongodump – db <db name

C:\Windows\System32\cmd.exe

```
Microsoft Windows [Version 10.0.19042.1826]
(c) Microsoft Corporation. All rights reserved.

D:\XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX\IIA Dataset and Assignment>mongodump --db INTENT
```

C:\Windows\System32\cmd.exe

```
2022-07-19T13:29:50.173+0530    writing INTENT.TblITSMPriorityMapping to dump\INTENT\TblITSMPriorityMappin
2022-07-19T13:29:50.174+0530    done dumping INTENT.TblImageAnalysis_defaultFlow (1 document)
2022-07-19T13:29:50.180+0530    writing INTENT.TblAssignmentEnableStatus to dump\INTENT\TblAssignmentEnabl
2022-07-19T13:29:50.181+0530    done dumping INTENT.TblRosterShifts (1 document)
2022-07-19T13:29:50.185+0530    writing INTENT.TblImageAnalysisColumns to dump\INTENT\TblImageAnalysisColu
2022-07-19T13:29:50.191+0530    done dumping INTENT.TblITSMPriorityMapping (1 document)
2022-07-19T13:29:50.193+0530    done dumping INTENT.TblAssignmentEnableStatus (1 document)
2022-07-19T13:29:50.196+0530    writing INTENT.TblDataset to dump\INTENT\TblDataset.bson
2022-07-19T13:29:50.204+0530    writing INTENT.TblITSMTools to dump\INTENT\TblITSMTools.bson
2022-07-19T13:29:50.206+0530    writing INTENT.TblPMAFields to dump\INTENT\TblPMAFields.bson
2022-07-19T13:29:50.207+0530    done dumping INTENT.TblImageAnalysisColumns (1 document)
2022-07-19T13:29:50.215+0530    done dumping INTENT.TblDataset (1 document)
2022-07-19T13:29:50.218+0530    done dumping INTENT.TblITSMTools (1 document)
2022-07-19T13:29:50.219+0530    writing INTENT.TblWorkflowTickets to dump\INTENT\TblWorkflowTickets.bson
2022-07-19T13:29:50.226+0530    done dumping INTENT.TblPMAFields (1 document)
2022-07-19T13:29:50.227+0530    writing INTENT.TblRPA_Configuration to dump\INTENT\TblRPA_Configuration.bs
2022-07-19T13:29:50.236+0530    writing INTENT.TblCluster to dump\INTENT\TblCluster.bson
2022-07-19T13:29:50.238+0530    writing INTENT.TblPredTicketsBotFactory to dump\INTENT\TblPredTicketsBotFa
2022-07-19T13:29:50.242+0530    done dumping INTENT.TblWorkflowTickets (1 document)
2022-07-19T13:29:50.244+0530    done dumping INTENT.TblRPA_Configuration (0 documents)
2022-07-19T13:29:50.251+0530    writing INTENT.TblManualAssignee to dump\INTENT\TblManualAssignee.bson
2022-07-19T13:29:50.257+0530    done dumping INTENT.TblCluster (0 documents)
2022-07-19T13:29:50.258+0530    done dumping INTENT.TblPredTicketsBotFactory (0 documents)
2022-07-19T13:29:50.262+0530    writing INTENT.TblApprovedTickets to dump\INTENT\TblApprovedTickets.bson
2022-07-19T13:29:50.263+0530    writing INTENT.TblNamedEntity to dump\INTENT\TblNamedEntity.bson
2022-07-19T13:29:50.264+0530    done dumping INTENT.TblManualAssignee (0 documents)
2022-07-19T13:29:50.268+0530    writing INTENT.TblRTTicketsBotFactory to dump\INTENT\TblRTTicketsBotFactor
2022-07-19T13:29:50.276+0530    done dumping INTENT.TblApprovedTickets (0 documents)
2022-07-19T13:29:50.280+0530    writing INTENT.TblRPADiagnostic to dump\INTENT\TblRPADiagnostic.bson
2022-07-19T13:29:50.287+0530    done dumping INTENT.TblNamedEntity (0 documents)
2022-07-19T13:29:50.288+0530    writing INTENT.TblKnownErrors to dump\INTENT\TblKnownErrors.bson
2022-07-19T13:29:50.292+0530    done dumping INTENT.TblRTTicketsBotFactory (0 documents)
2022-07-19T13:29:50.298+0530    writing INTENT.TblDiagnosticScripts to dump\INTENT\TblDiagnosticScripts.bs
2022-07-19T13:29:50.300+0530    done dumping INTENT.TblRPADiagnostic (0 documents)
2022-07-19T13:29:50.304+0530    done dumping INTENT.TblKnownErrors (0 documents)
2022-07-19T13:29:50.310+0530    writing INTENT.TblImagePrediction to dump\INTENT\TblImagePrediction.bson
2022-07-19T13:29:50.311+0530    done dumping INTENT.TblDiagnosticScripts (0 documents)
2022-07-19T13:29:50.318+0530    done dumping INTENT.TblImagePrediction (0 documents)
```

OneDrive - Infosys Limited > IIA Dataset and Assignment > dump > INTENT

Name	Date modified	Type	Size
TblAlgorithm.bson	7/19/2022 1:29 PM	BSON File	2 KB
TblAlgorithm.metadata.json	7/19/2022 1:29 PM	JSON File	1 KB
TblApplication.bson	7/19/2022 1:29 PM	BSON File	3 KB
TblApplication.metadata.json	7/19/2022 1:29 PM	JSON File	1 KB
TblApplicationAnalystMapping.bson	7/19/2022 1:29 PM	BSON File	28 KB
TblApplicationAnalystMapping.metadata.json	7/19/2022 1:29 PM	JSON File	1 KB
TblApprovedTickets.bson	7/19/2022 1:29 PM	BSON File	0 KB
TblApprovedTickets.metadata.json	7/19/2022 1:29 PM	JSON File	1 KB

Restore:

Open command prompt outside dump folder where you kept **mongodump** files

mongorestore --db <databasename> --drop dump/<databasename>

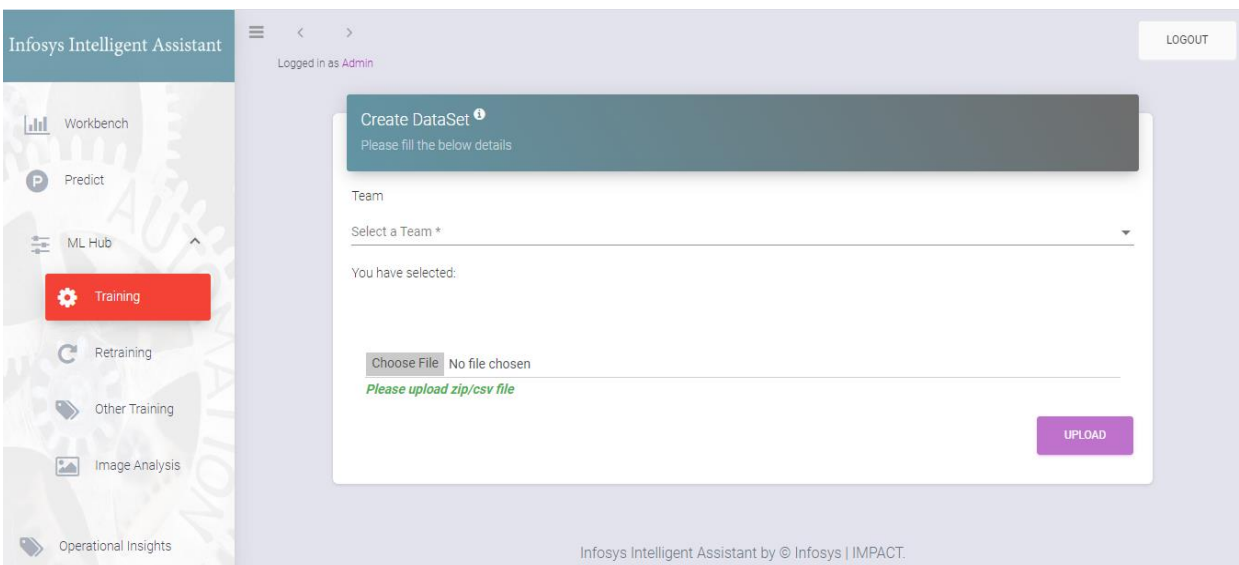
```
IIA Dataset and Assignment>mongorestore --db INTENT --drop dump/INTENT
```

T

1. IIA Dataset Setup

Uploading dataset for training in IIA:

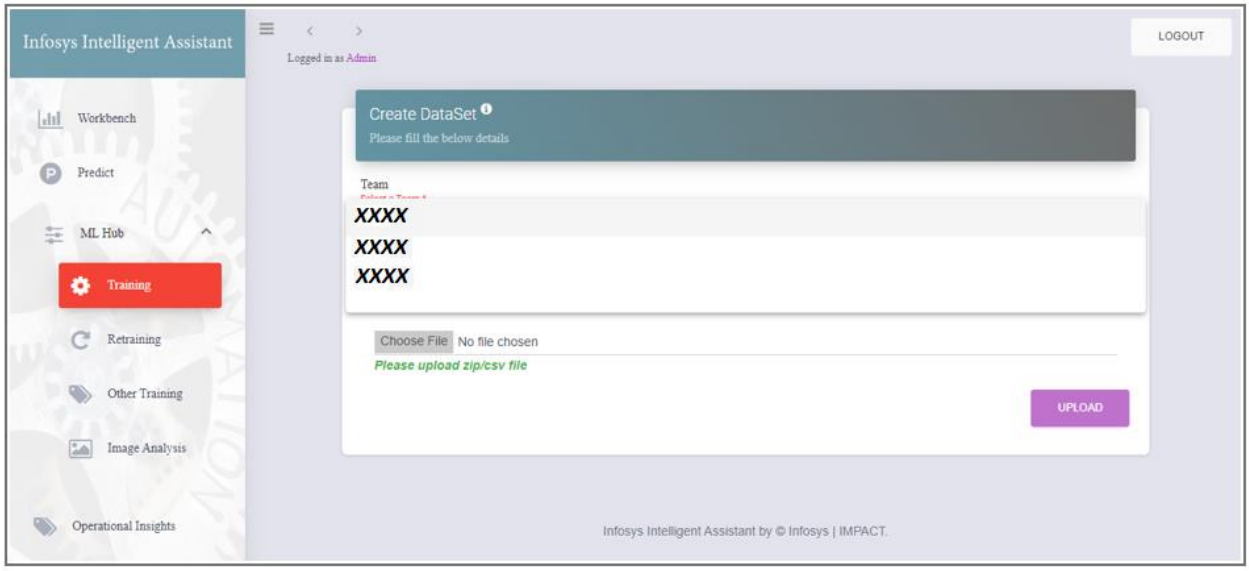
Below is screenshot of Training Module



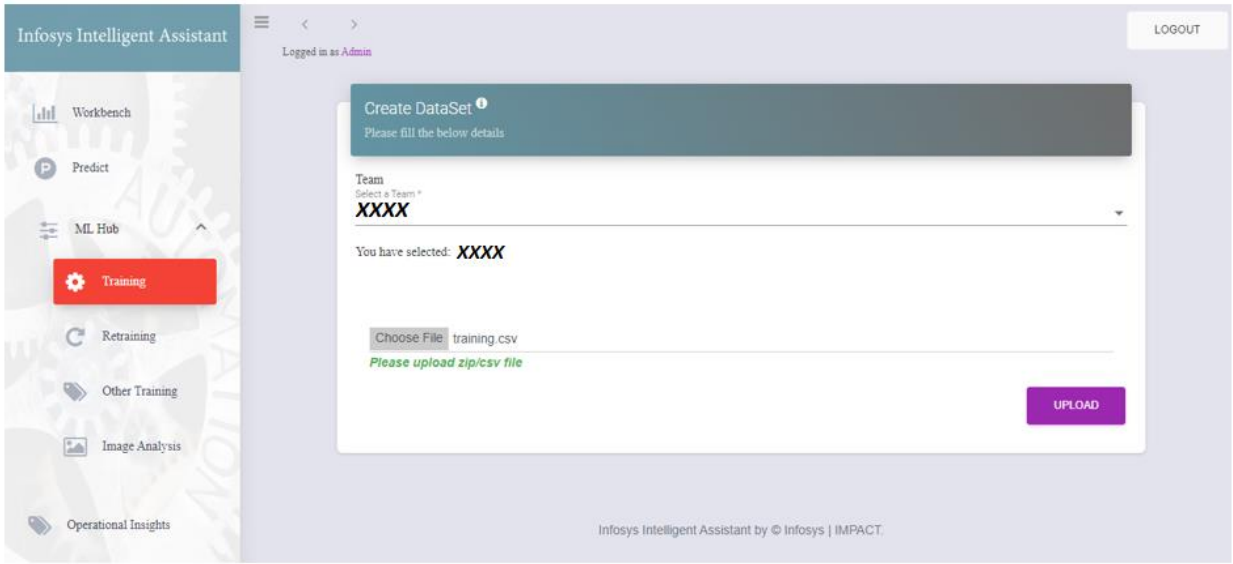
To add records, you will have to choose the team and file(zip/csv) to be uploaded.

Steps to add Records:

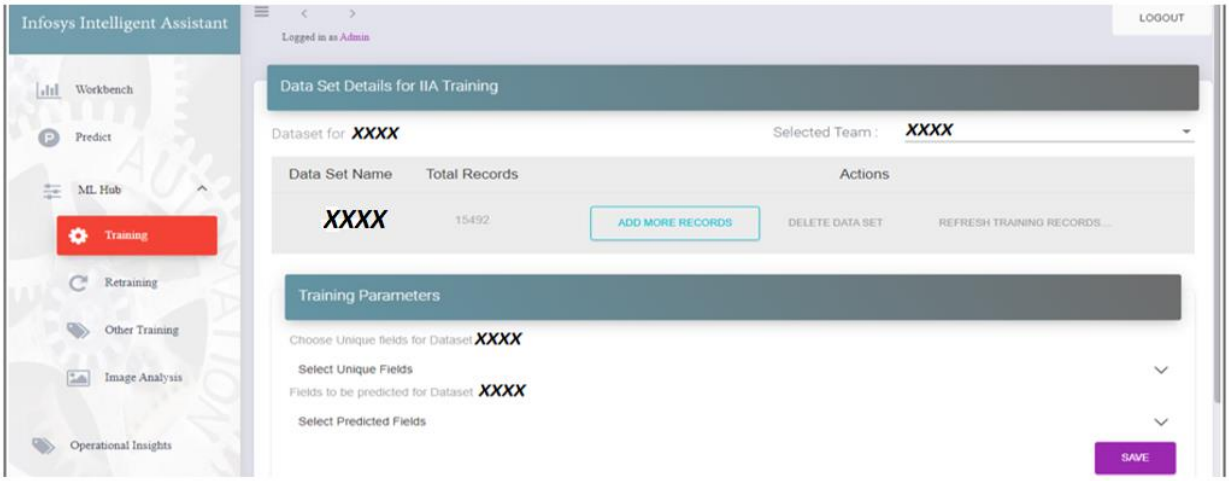
Step 1: Hover over the Select a Team option. Select the team you want to add the training data for, from the drop down.



Step 2: After Selecting the Team, choose the CSV/Zip file which you want to upload.



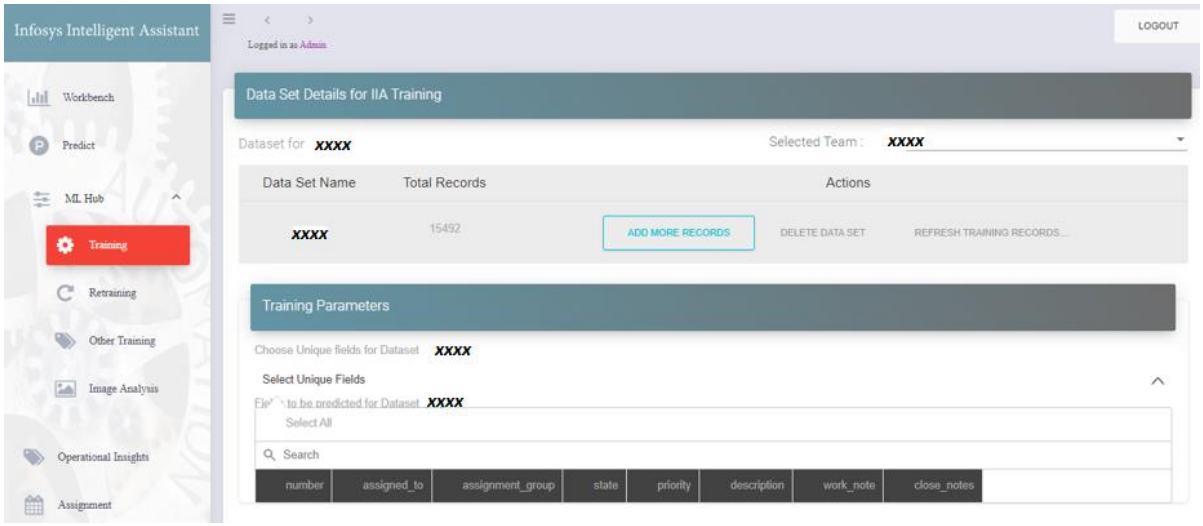
Step 3: Click on upload to upload the csv to IIA database. The tickets which have been uploaded are not trained yet.



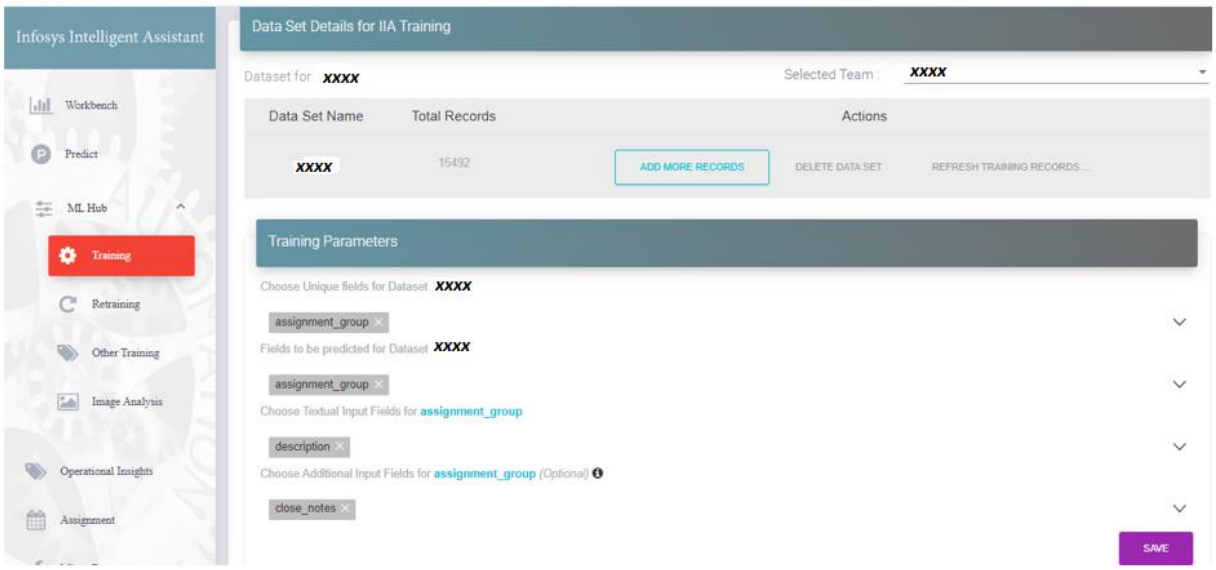
After you upload the records for training, you get an option to add more records or delete the dataset or refresh the training records added

Steps to select the Training Parameters:

Step 1: Click on Select Unique field to select the unique field from the uploaded tickets.

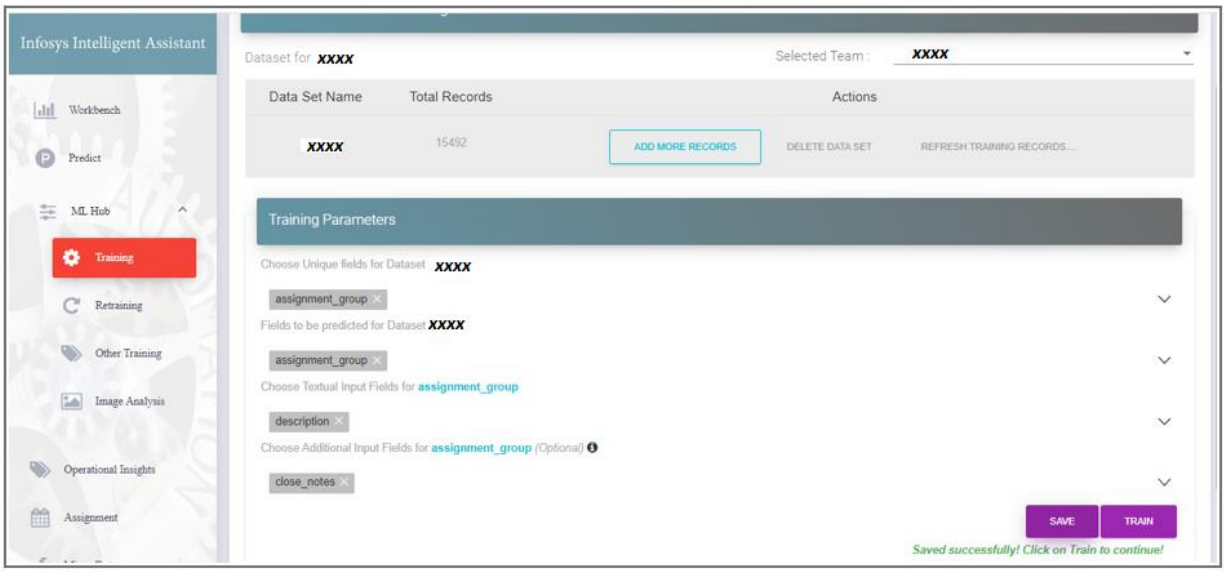


Step 2: Click on Select Predicted Fields to select the various field which you want to predict from IIA.
Step 3: Click on Select Textual Input Fields to choose input field for all the predicted Field.
Step 4: Click on Select Additional Input Fields to choose input field for all the predicted Field. The additional fields are optional.

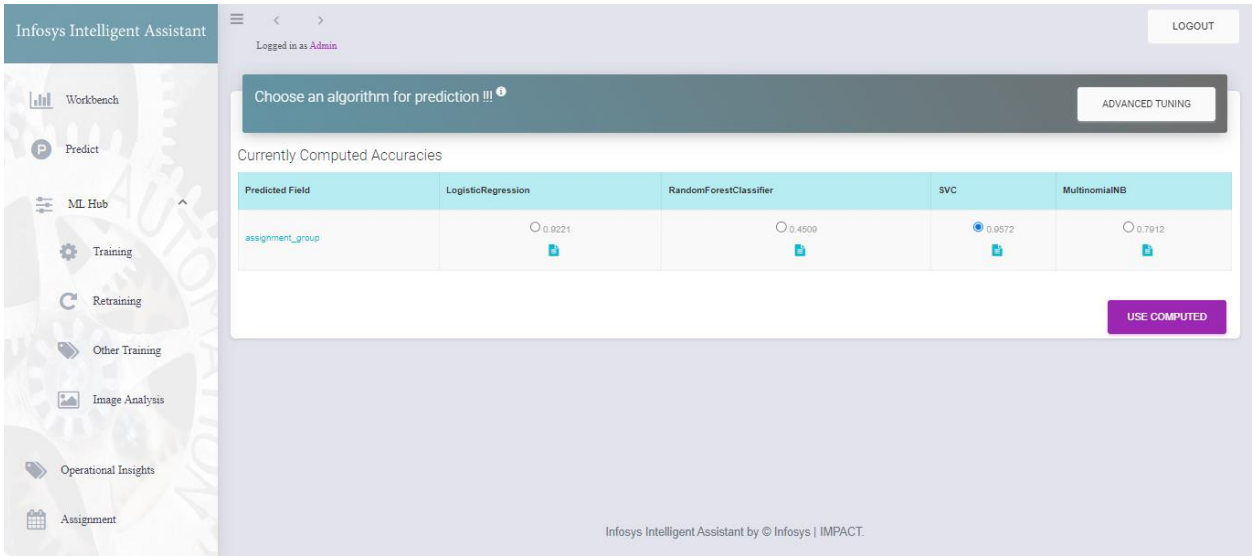


Step 5: Click on Save to save the parameter choices.

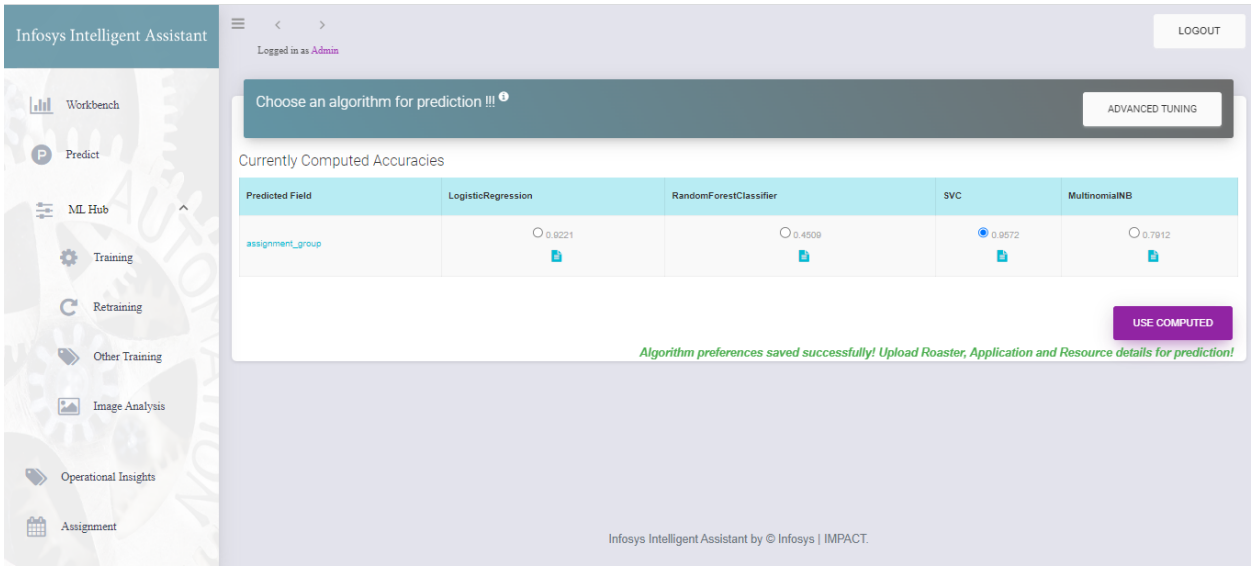
Step 6: Click on train to start the machine training on the selected parameters.



After the machine gets Trained on the chosen training parameter, IIA displays F1-score on the Training data for each Algorithm. By default, the algorithm giving the highest F1-score for that Predicted field is already selected by IIA. However, one can change the algorithm to be used by selecting a different radio button.



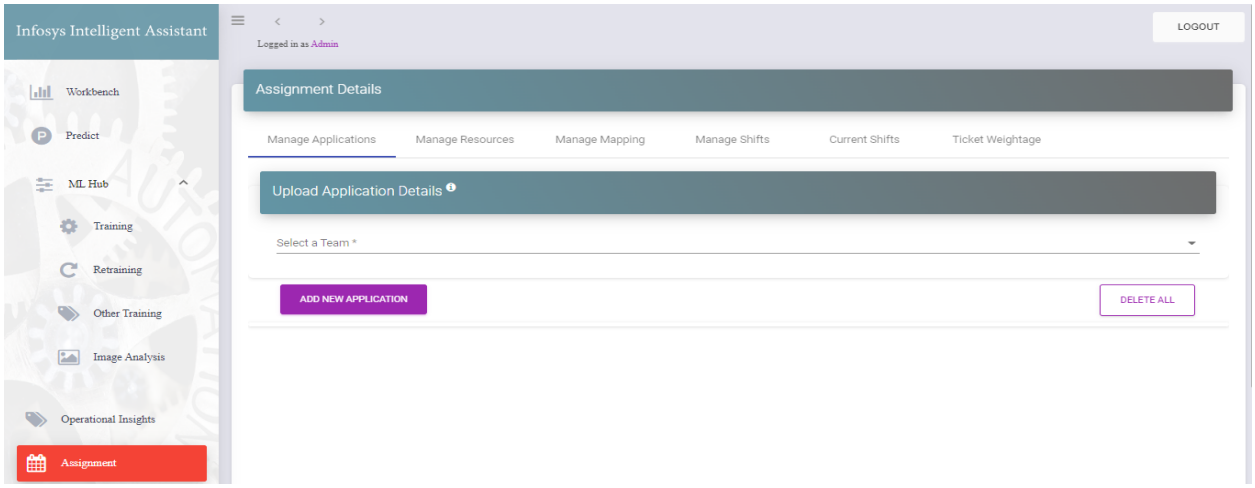
Step 7: After the algorithm to be used is selected the Button of USE COMPUTED must be pressed to save the chosen algorithms for future prediction use.



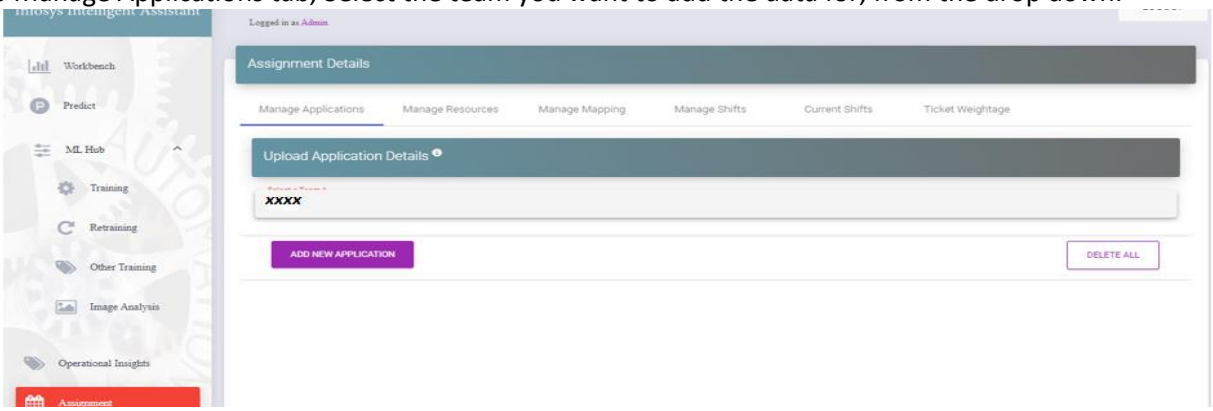
Now you can go ahead and upload supporting documents in Assignment module for prediction.

Steps to upload supporting documents in Assignment module :

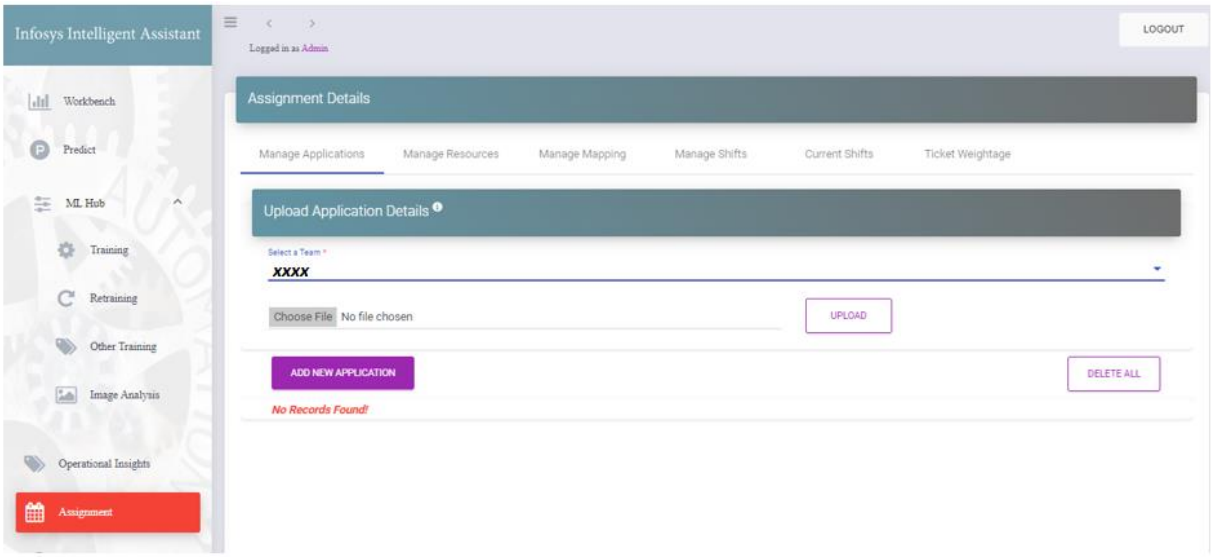
Below is screenshot of Assignment Module



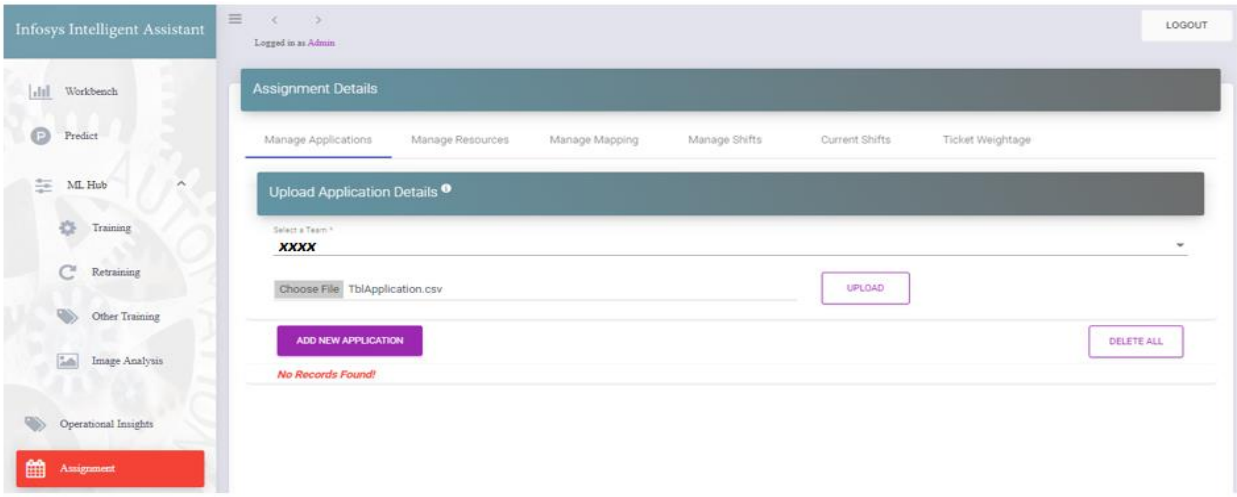
1. Go to Manage Applications tab, Select the team you want to add the data for, from the drop down.



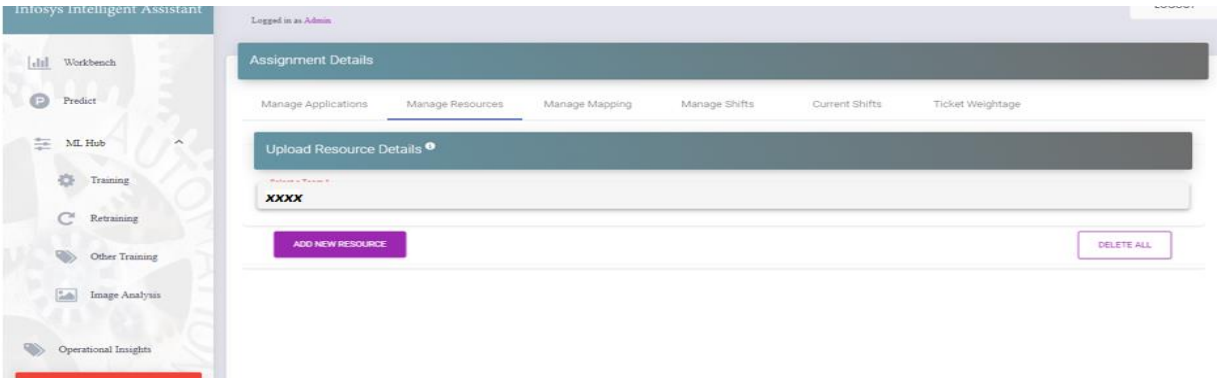
2. Choose the csv file here which has relevant information about the assignment groups or applications.



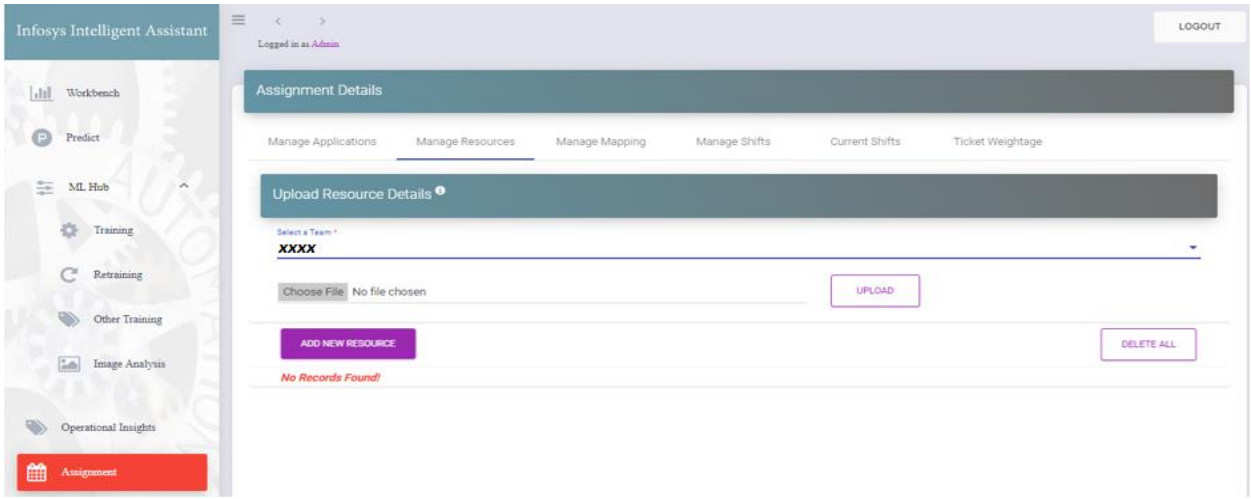
3. Click on Upload to upload csv data.



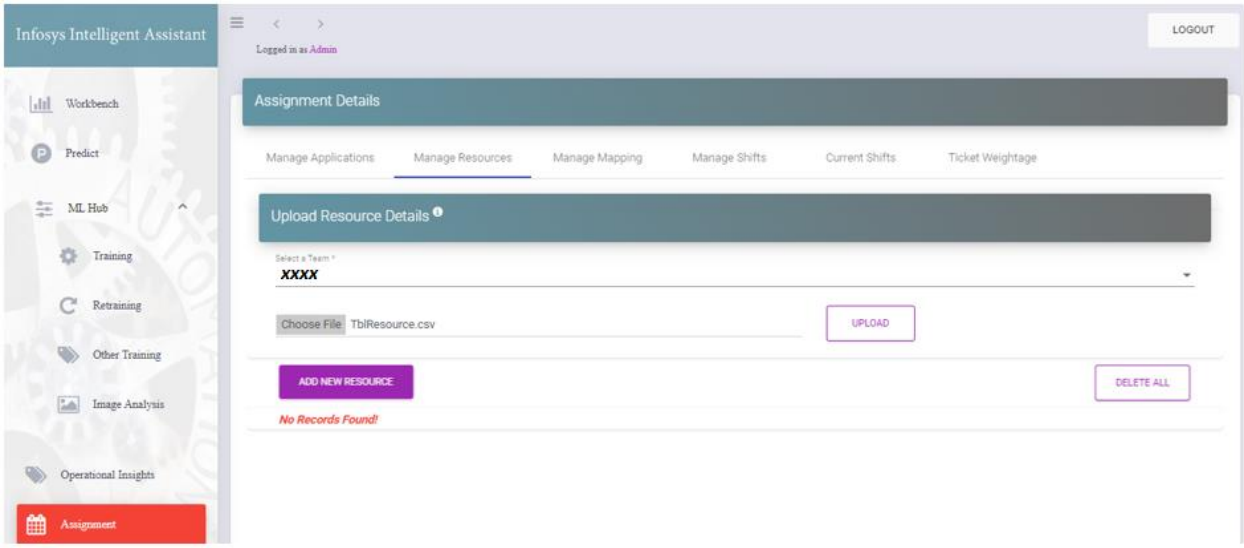
1. Click on Manage Resources tab, Select the team you want to add the data for, from the drop down.



2. After Selecting the Team, choose the csv file which contains the work details about a resource.



3.Click on Upload to upload csv data.



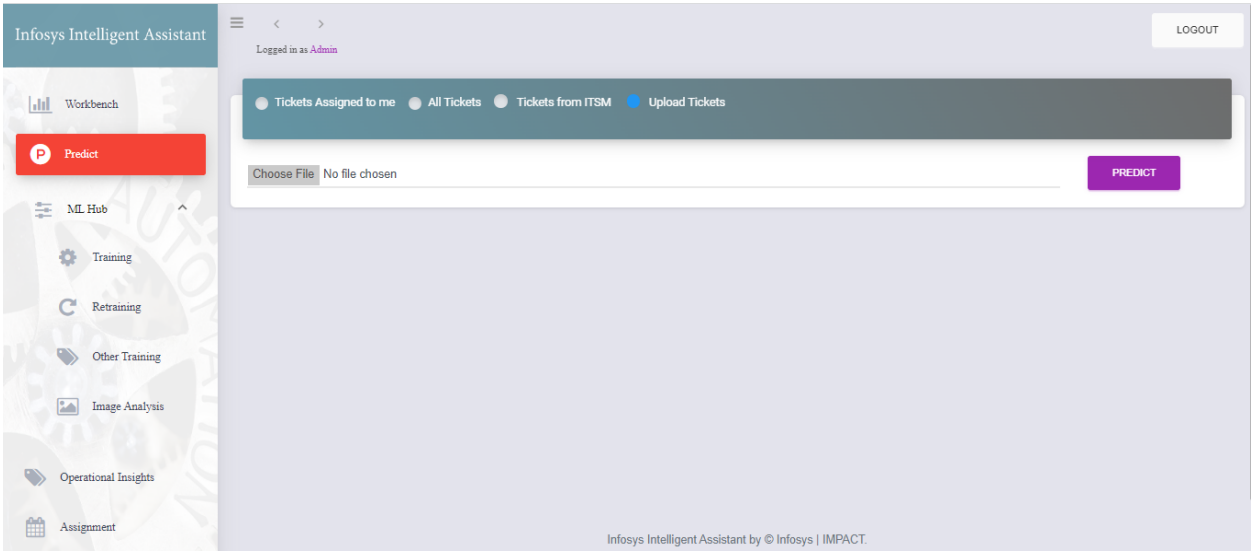
Then go to Manage Shift tab, select the team and upload the excel file which contains the shift details of the resource.

The last component of the module is the Ticket weightage. Here, select the team and upload a ticket details via csv file, such as Ticket type, priority, status, weightage details.

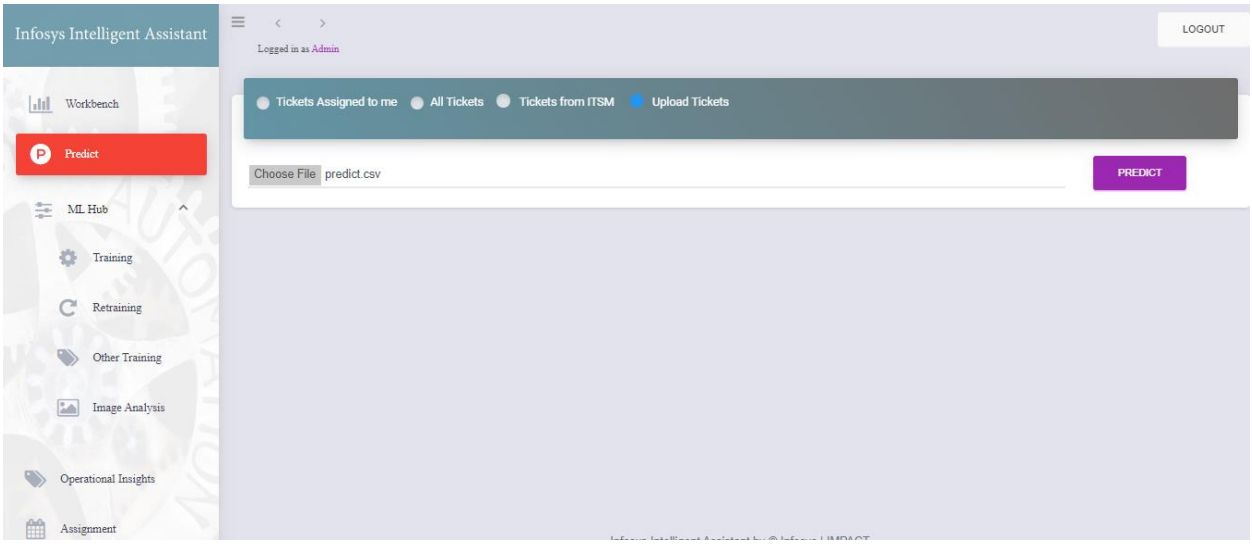
Once we have uploaded all the relevant files here, we can now proceed with the prediction of the assignee for the tickets.

Uploading data set for Prediction:

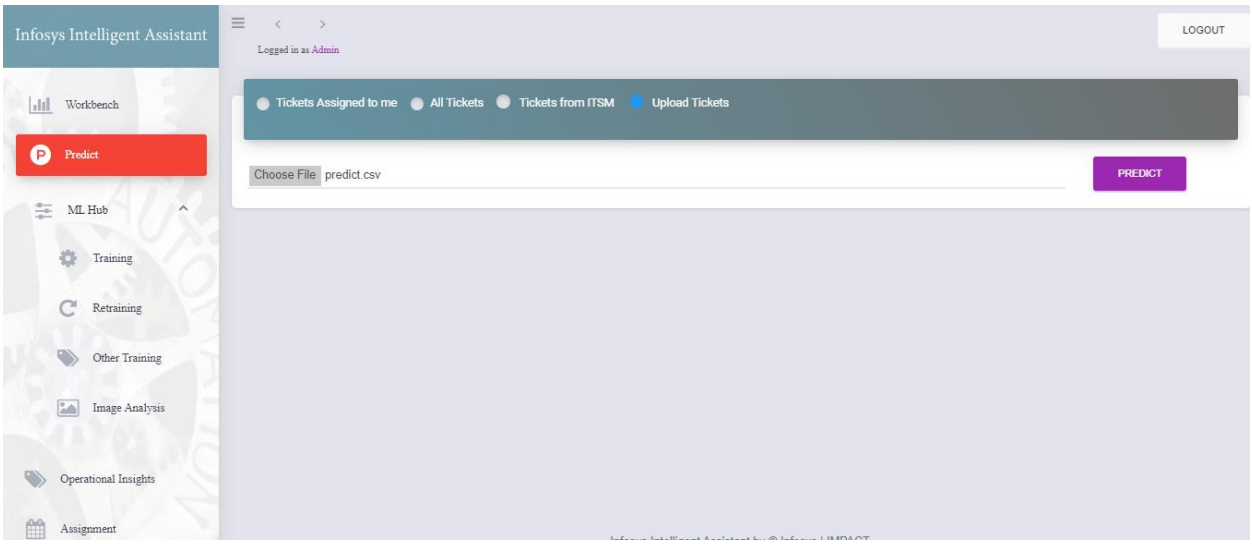
Below is screenshot of Prediction Module



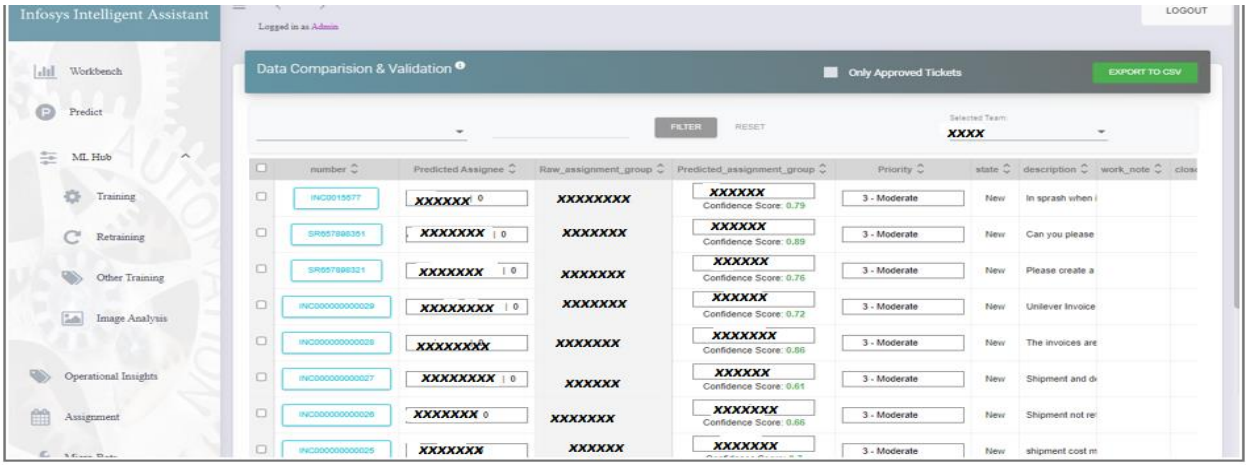
Step 1: Here choose the csv file to upload for prediction.



Step 2: Click on Predict button to start prediction for the tickets.

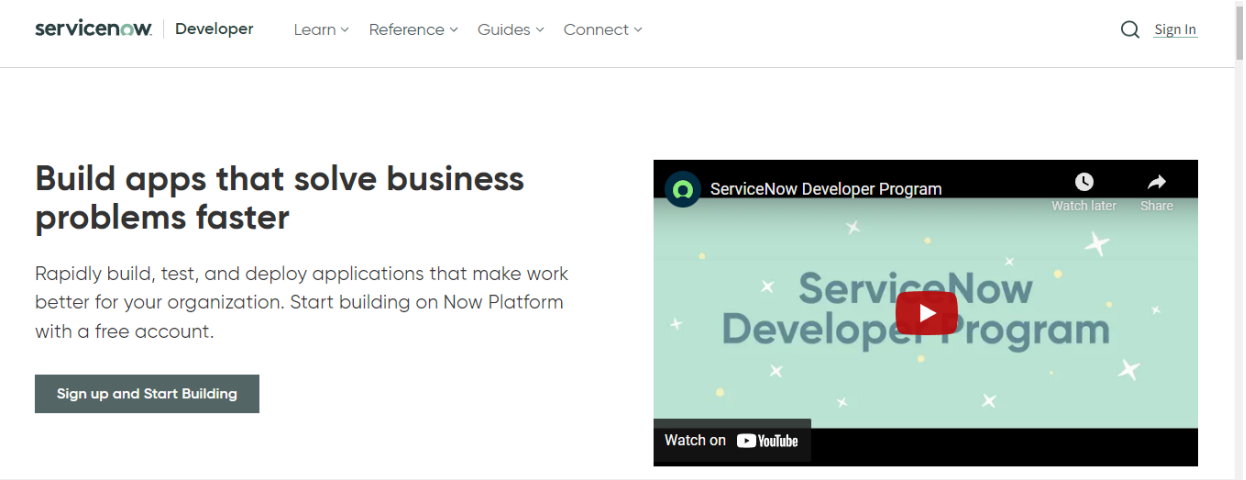


Once the tickets given for prediction have been Predicted, A Data Comparison & Validation Page Opens. In this page you can see the various ticket details as per uploaded dataset along with Ticket Number, Predicted Assignee, Raw assignment group and Predicted assignent group.

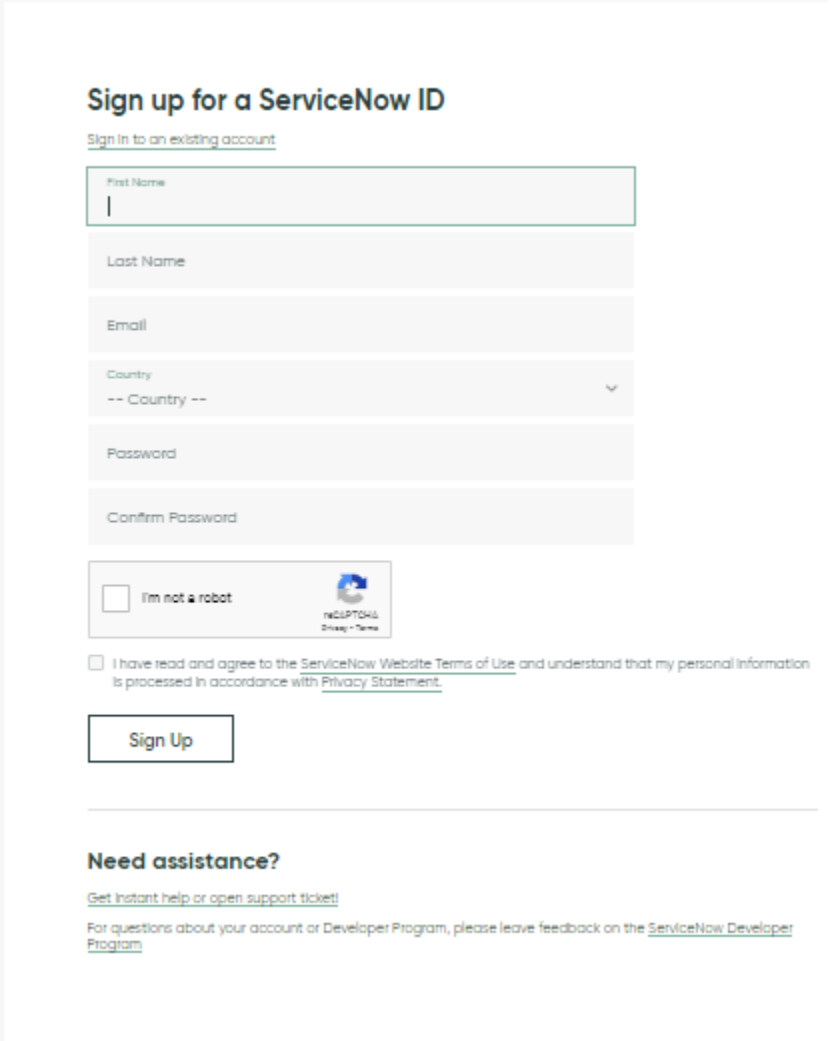


3. Service Now Setup

Step 1 : Go to https://developer.servicenow.com/dev.do



Step 2 : Click on "**Sign up and start building** ". you will be navigated to sign up page enter all the required details and clik on sign up.



Once it is Done you will get get below message

Thank You!

A ServiceNow ID has been created for you with ***xxxxxxxxxxxxxx***@gmail.com. This application requires you to verify your email. Please check your email for a link to verify your account. After verification you can login to this application.

Please Note: The email will be valid for 24 hours.

You will receive a mail like shown below click on verify Email.

Hello ***XXXXX*** ,

You recently registered with ServiceNow - Developer Program. You are receiving this email so we can confirm this email address for your ServiceNow ID.

Please click below to verify your email address:

Verify Email

If clicking on the above button does not work, please copy and paste the URL below in a browser to verify your email address:

https://signon.service-now.com/x_snc_sso_auth.do#/

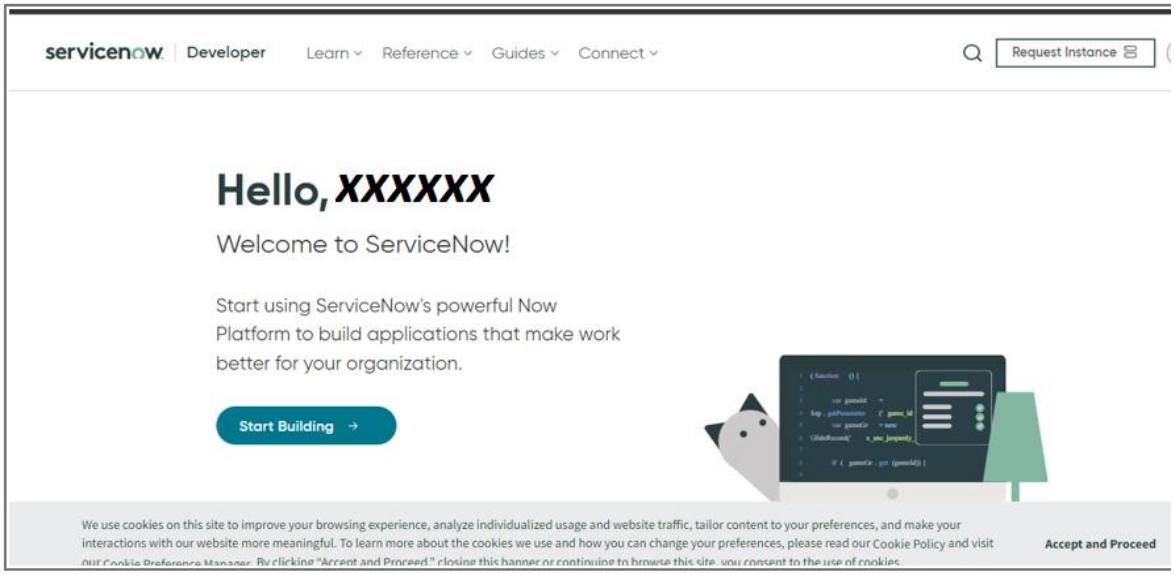
You will get this verification completion mail.

Thank You!

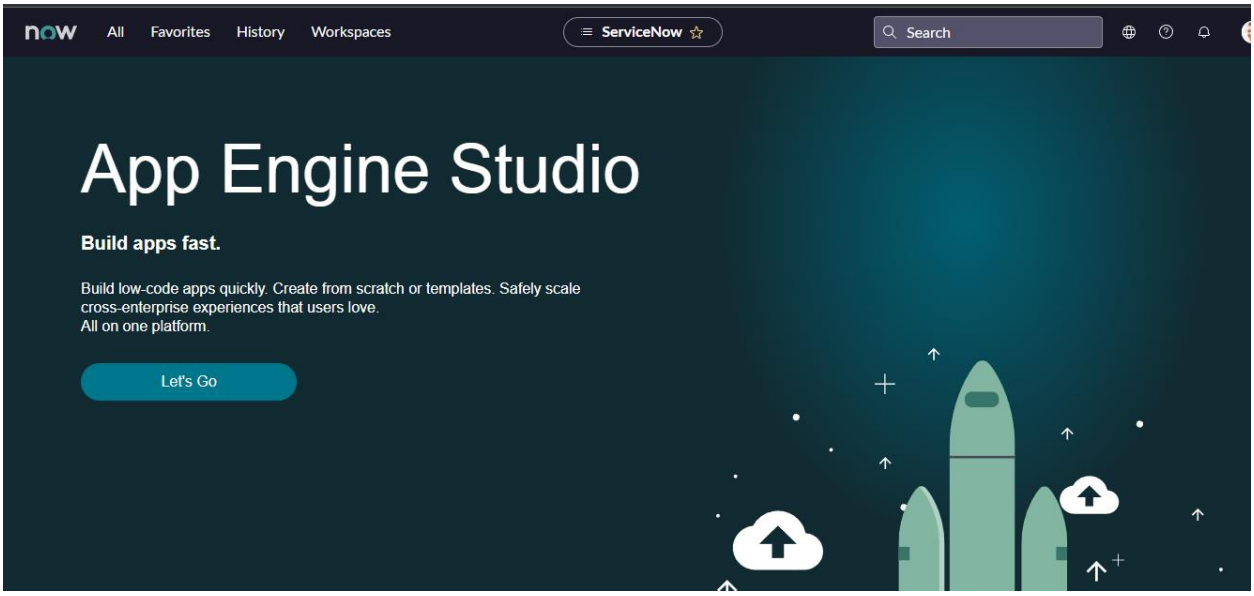
Your email: ***XXXXXXXXX*** @gmail.com has been verified. Please click below to login to the application.

Login

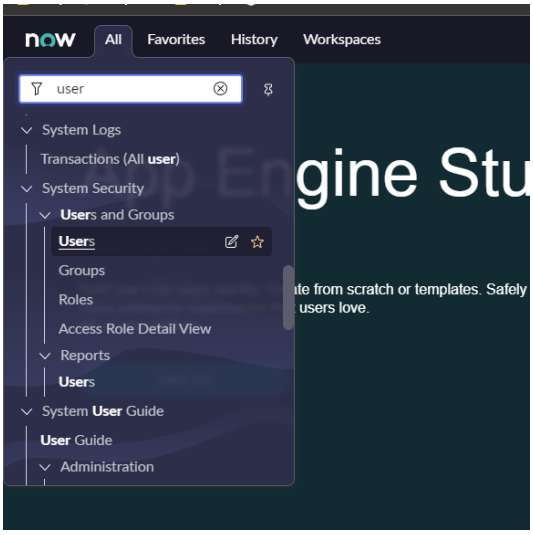
Click on **login** and give the required credentials.



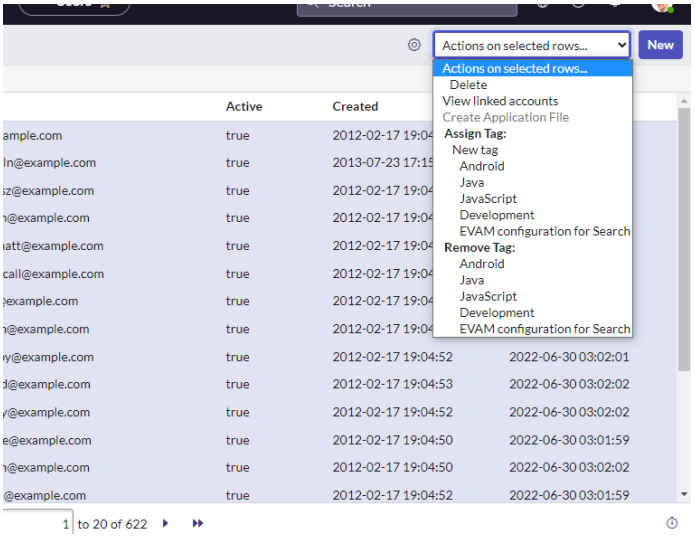
Click on **"request Instance"** once it is completed click on start building this page will be shown.



Step 1 : Click on All :



Step 2 : Remove All previous data

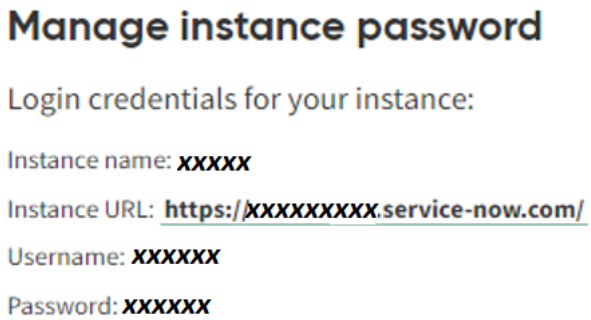


Uploading the Data :

1. **Uploading user details :**
   for uploading tickets search for user click on the one which is inside system security.
   To upload the new users first prepare the file update these fields email,firstName,lastName , user id , password fields and upload the user in Service Now.
   Right click on user id > click on import > choose the file location > click on upload > click on preview imported data (validate if all got uploaded) > click on complete Import .
2. **To upload assignment group :**
   for uploading tickets search for group click on the one which is inside system security.
   To upload the assigment groups first update these fields name and description in sys_user_group_insert file.
   Right click on Name> click on import > choose the file location > click on upload > click on preview imported data (validate if all got uploaded) > click on complete Import .
3. **To upload tickets in bulk :**
   for uploading tickets search for incident click on the one which is inside service desk.
   To upload the tickets first update the required fields use the template file.
   Right click on Number > click on import > choose the file location > click on upload > click on preview imported data (validate if all got uploaded) > click on complete Import .

To integrate it with IIA :

Now go to your profile click on Manage instance password



Update your credentials in mongodb campass in TblITSMDetails collection.

Key	Value	Type
(1) ObjectId("5c90869e22571b30106eca69")	{ 8 fields }	Object
_id	ObjectId("5c90869e22571b30106eca69")	ObjectId
CustomerID	1	Int32
UserID	XXXXXX	String
password	XXXXXX	String
ITSMToolName	SNOW	String
MEUrl	https://XXXXXservice-now.com/	String
MeAuthToken		String
ITSMInstance	XXXXXX	String

4. IIA Deployment

Build Config:

Modify the Version and build as required in build config so that it will reflect in intent wheel package

build_config.yaml

```yaml
-
    name: intent
    version: 3.0.0
    build: 0.0.1
    author: XXXXXXXXX
    author_email: XXXXXXXXXXXXXXXX@infosys.com
    description: Infosys Intelligent Assistant
    long_description: Infosys Intelligent Assistant
    classifiers: ["Programming Language :: Python :: 3",
                  "License :: OSI Approved :: MIT License",
                  "Operating System :: OS Independent",]
    package_dir: {"": "src"}
    packages: src
    python_requires: ['>=3.6']

-
    name: intent_custom
    version: 3.0.0
    build: 0.0.1
    author: XXXXXXX
    author_email: XXXXXXXXXXXXXXXX@infosys.com
    description: Infosys Intelligent Assistant
    long_description: Infosys Intelligent Assistant
    classifiers: ["Programming Language :: Python :: 3",
                  "License :: OSI Approved :: MIT License",
                  "Operating System :: OS Independent",]
    package_dir: {"": "account_specific"}
    packages: account_specific
    python_requires: ['>=3.6']
```

Open command prompt and issue **set PYTHONPATH=src & python create_wheel_file.py** to create intent wheel file package for deployment. Package will get created inside **dist** folder upon succesful build.

This PC > DATA (D:) > Github > Development > intent-python-dev > dist

Name	Date modified	Type	Size
intent-3.0.0_build_0.0.1-py3-none-any.whl	7/19/2022 1:51 PM	WHL File	411 KB
intent-3.0.0-build-0.0.1.tar.gz	7/19/2022 1:50 PM	GZ File	273 KB

Deploy build file:

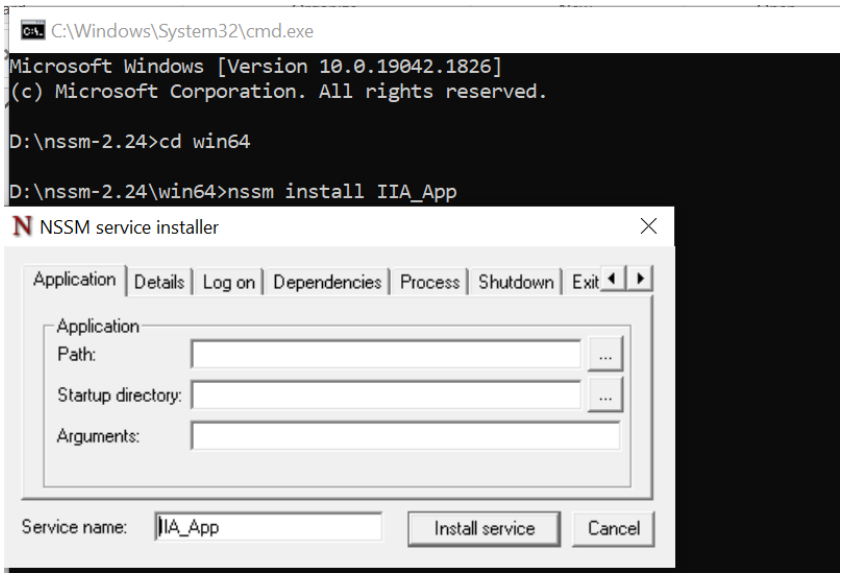
Install the same in client/deployment environment to deploy the code on issuing

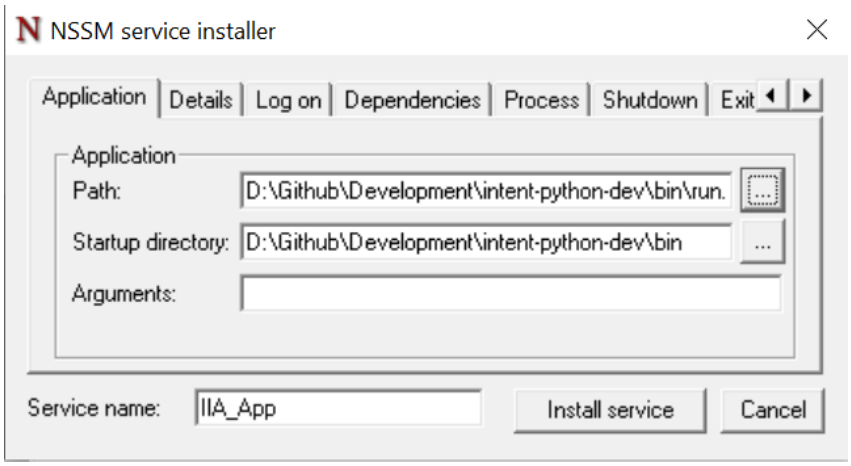
pip install intent-3.0.0_build_0.0.1-py3-none-any.whl in command prompt

5. IIA as Service

Download **nssm package** and extract the zip. Place it either in C drive or D drive according to the system drive

Open command prompt and follow the steps below:





Click **Install** Service and **start** the service under **services.msc**